

Task 12 – Missing Value Identification

Project Report: Titanic Dataset Missing Value Analysis

1. Objective

The objective was to identify missing values in the Titanic dataset, calculate their percentages, summarize the results, and visualize the columns containing missing data.

2. Dataset

Dataset: Titanic Dataset

Total Records: 891

Total Columns: 15

Tools: Microsoft Excel, Python, Pandas, Matplotlib, and Google Colab

3. Work Performed

- Loaded the Titanic CSV dataset into Python using Pandas.
- Checked every column for missing values using `df.isnull().sum()`.
- Calculated missing-value percentages using `df.isnull().mean() * 100`.
- Created a missing-value summary table in Excel.
- Created a bar chart using Matplotlib for columns containing missing values.
- Documented the findings in a GitHub README.

4. Missing Value Summary

Column	Missing Values	Missing %
deck	688	77.22%
age	177	19.87%
embarked	2	0.22%
embark_town	2	0.22%
Other columns	0	0.00%

5. Excel Analysis

A separate sheet named “Missing Value Summary” was created. COUNTBLANK was used to identify blank cells, ROWS was used to determine the 891 records, and the missing percentage was calculated as Missing Values / Total Records.

6. Python Analysis

Main commands used:

```
df.isnull().sum()
```

```
df.isnull().mean() * 100
```

A Pandas summary DataFrame was created and the missing values were visualized with a Matplotlib bar chart.

7. Key Findings

- The dataset contains 891 passenger records and 15 columns.
- The deck column has 688 missing values (77.22%).
- The age column has 177 missing values (19.87%).
- The embarked and embark_town columns each have 2 missing values (0.22%).
- The remaining columns contain no missing values.
- Missing values should be handled appropriately rather than removed without justification.

8. Conclusion

The task was completed by identifying, quantifying, summarizing, and visualizing missing values in the Titanic dataset. Excel was used for tabular analysis, while Python with Pandas and Matplotlib was used for programmatic analysis and visualization. The results were documented in GitHub.

9. Project Files

- titanic(1).csv – Dataset
- Titanic_Task12_Missing_Value_Analysis.ipynb – Python analysis
- Titanic_Task12_Missing_Values.xlsx – Excel analysis
- README.md – GitHub project documentation